SOFTWARE REQUIREMENTS SPECIFICATION COLLABORATIVE STUDY MATERIAL & QUIZ PLATFORM

Version: 1.0

Programming Language: Python

Team Members

1. Neha Prabhu - PES2UG24CS312
2. Nisarga - PES2UG24CS325
3. Pradhyumn Anand - PES2UG24CS358
4. Neeraj R Rugi - PES2UG24CS311

Revision History

Version	Date	Author(s)	Change Summary
1.0	03-09-2026	Project Team	Initial SRS

Table of Contents

1. Introduction

2. Overall Description

3. External Interface Requirements

4. System Features (Detailed)

5. Non-Functional Requirements (Detailed)

5.1 Security

6. Quality Attributes & Acceptance Tests

7. UML Use-Case Diagrams

8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional and security requirements of the Collaborative Study Material & Quiz Platform. The system provides a centralized environment for sharing study materials, maintaining a question bank, creating and attempting quizzes, and analysing learner performance.

1.2 Scope

The platform covers user authentication, study-material upload and sharing, material search and access, question-bank management, quiz creation, quiz attempts and automatic evaluation, result history, and performance analysis. It is intended for academic use and does not cover online proctoring, live video classes, or payment processing.

1.3 Audience

Students, instructors/contributors, administrators, developers, testers and project evaluators.

1.4 Definitions

Study Material , A digital academic resource such as notes, documents, presentations or reference files shared through the platform.

Question Bank , A collection of questions stored and organized for reuse in quizzes.

Quiz , A set of questions published for a learner to attempt within the platform.

Contributor , An instructor or authorized user who can upload materials and create/manage questions or quizzes.

Performance Analysis , Reports and summaries derived from quiz attempts, scores and question-wise results.

2. Overall Description

2.1 Product Perspective

The platform is a web-based application with a Python backend and persistent database. It replaces scattered sharing of notes and manually maintained quiz records with a single system in which academic resources, questions, quizzes and learner results are connected.

2.2 Major Product Functions

- User registration, login and role-based access
- Study-material upload, categorization, search and access
- Question-bank creation, editing and organization
- Quiz creation and publication
- Quiz attempt, submission and automatic scoring
- Result history and performance analysis
- Administrative user and content management

2.3 User Roles and Characteristics

Role	Main Capabilities
Student	Browse/share permitted materials, search resources, attempt quizzes and view personal results and performance.
Contributor / Instructor	Upload and manage study materials, create/manage questions and quizzes, and view relevant learner results.
Administrator	Manage users, roles and inappropriate content; maintain overall platform records.

2.4 Operating Environment

- Standard web browser on desktop or mobile device
- Python-based application server
- Relational database for persistent data
- Windows or Linux development/deployment environment

2.5 Constraints

- Only authenticated and authorized users shall perform protected operations.
- Uploaded materials and question content must satisfy configured file, size and validation rules.
- Quiz scoring depends on correctly configured questions and answers.
- The platform requires access to the application server and database for normal operation.

3. External Interface Requirements

3.1 User Interfaces

- Login/registration page
- Student dashboard
- Material library and material upload page
- Question-bank management page
- Quiz creation and attempt page
- Results and performance dashboard
- Administrator dashboard

3.2 Hardware Interfaces

No specialized hardware is required. The system shall operate on a standard computer, tablet or mobile device with network access.

3.3 Software Interfaces

The frontend shall communicate with the Python backend using standard web requests. The backend shall communicate with a relational database for users, materials, questions, quizzes, attempts and results. File storage shall be used for uploaded study materials.

3.4 Communication Interfaces

The application shall use HTTP/HTTPS for client-server communication, with HTTPS used for production deployment.

4. System Features

Each functional requirement includes acceptance criteria and a test-case reference. Requirement IDs follow the format CSMQ-F-###.

4.1 Authentication & User Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
CSMQ-F-001	The system shall allow a registered user to log in using a valid email/username and password.	High	AC: Valid credentials create an authenticated session; invalid credentials show an error. Test: TC-Auth-01	All roles
CSMQ-F-002	The system shall allow an authenticated user to log out and terminate the current session.	Medium	AC: Session is terminated and protected pages are inaccessible. Test: TC-Auth-02	-
CSMQ-F-003	The system shall assign users a role and restrict protected functions according to that role.	High	AC: A Student cannot access Contributor/Admin functions; authorized roles can access permitted functions. Test: TC-Auth-03	RBAC

4.2 Study Material Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
CSMQ-F-004	The system shall allow authorized users to upload study materials with title, subject/category and description.	High	AC: Valid material is stored and appears in the contributor's material list. Test: TC-Mat-01	Authentication required
CSMQ-F-005	The system shall allow users to browse and search available study materials using fields such as title, subject or category.	High	AC: Search returns only matching accessible materials. Test: TC-Mat-02	-
CSMQ-F-006	The system shall allow authorized contributors to update or remove materials they manage.	Medium	AC: Changes are reflected in the library and removed material is no longer available to users. Test: TC-Mat-03	Ownership check
CSMQ-F-007	The system shall allow users to view or download study materials that they are authorized to access.	High	AC: A permitted material opens/downloads successfully; unauthorized access is denied. Test: TC-Mat-04	Access control

4.3 Question Bank & Quiz Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
CSMQ-F-008	The system shall allow authorized contributors to create questions with question text, options, correct answer and topic.	High	AC: A valid question is stored and can be retrieved from the question bank. Test: TC-QB-01	-
CSMQ-F-009	The system shall allow contributors to edit or delete questions that they manage.	Medium	AC: Updated question data is saved and deleted questions are unavailable for new quizzes. Test: TC-QB-02	Ownership check
CSMQ-F-010	The system shall allow contributors to create a quiz by selecting questions from the question bank and defining a title and basic instructions.	High	AC: A quiz is created with the selected questions and metadata. Test: TC-Quiz-01	Depends on F-008
CSMQ-F-011	The system shall allow a contributor to publish or unpublish a quiz.	Medium	AC: Published quizzes are available to permitted students; unpublished quizzes are not available	-

			for new attempts. Test: TC-Quiz-02	
CSMQ-F-012	The system shall allow a student to view the questions of an available quiz and submit answers.	High	AC: The attempt records the selected answers and submission time. Test: TC-Quiz-03	Published quiz

4.4 Results & Performance Analysis

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
CSMQ-F-013	The system shall automatically evaluate submitted objective quiz answers and calculate the score.	High	AC: Score equals the number of correct answers according to the configured answer key. Test: TC-Result-01	-
CSMQ-F-014	The system shall store each completed quiz attempt with student, quiz, score and submission details.	High	AC: A completed attempt is present in the student's result history. Test: TC-Result-02	-
CSMQ-F-015	The system shall allow students to view their quiz scores and performance history.	Medium	AC: The dashboard shows previous attempts and corresponding scores. Test: TC-Result-03	-
CSMQ-F-016	The system shall provide performance summaries such as average score, quiz-wise performance and topic-wise performance where topic data is available.	Medium	AC: Summary values are calculated from stored attempts and displayed correctly. Test: TC-Perf-01	Requires result data

4.5 Administration

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
CSMQ-F-017	The system shall allow an administrator to view, activate or deactivate user accounts.	Medium	AC: Account status changes are enforced on subsequent login/access. Test: TC-Admin-01	Admin only
CSMQ-F-018	The system shall allow an administrator to review and remove inappropriate or invalid shared content.	Medium	AC: Removed content is no longer accessible through normal library/search functions. Test: TC-Admin-02	Admin only

5. Non-Functional Requirements

Req ID	Requirement	Category	Priority	Acceptance / Measurement
CSMQ-NF-001	Common operations such as login, material search and result viewing shall complete within 3 seconds for 90% of requests under normal test load.	Performance	High	90th percentile response time ≤ 3 s. Test: TC-Perf-01
CSMQ-NF-002	The system shall maintain data consistency between questions, quizzes, attempts and calculated results.	Data Integrity	High	Stored result matches the submitted answers and configured answer keys in all test cases. Test: TC-INT-01
CSMQ-NF-003	The interface shall use a consistent layout and clear navigation so that a first-time user can locate material and attempt an available quiz without external help.	Usability	Medium	Usability walkthrough completed successfully. Test: TC-UX-01
CSMQ-NF-004	The application shall be organized into modular components for authentication, materials, question bank, quizzes and analytics.	Maintainability	Medium	Modules can be modified with minimal unrelated changes. Test: Code Review
CSMQ-NF-005	The system shall be accessible through the two most recent versions of major desktop browsers.	Compatibility	Medium	Core workflows function on supported browsers. Test: TC-Compat-01
CSMQ-NF-006	The system shall remain available during normal operating periods excluding scheduled maintenance.	Reliability	Medium	No unplanned failure during the defined acceptance monitoring period. Test: Operations Review

5.1 Security

5.1.1 Security Objectives

- Protect user accounts, uploaded academic content and performance records from unauthorized access.
- Ensure that only authorized roles can modify quizzes, questions, user accounts and shared content.
- Maintain confidentiality and integrity of stored credentials and application data.

5.1.2 Security Requirements

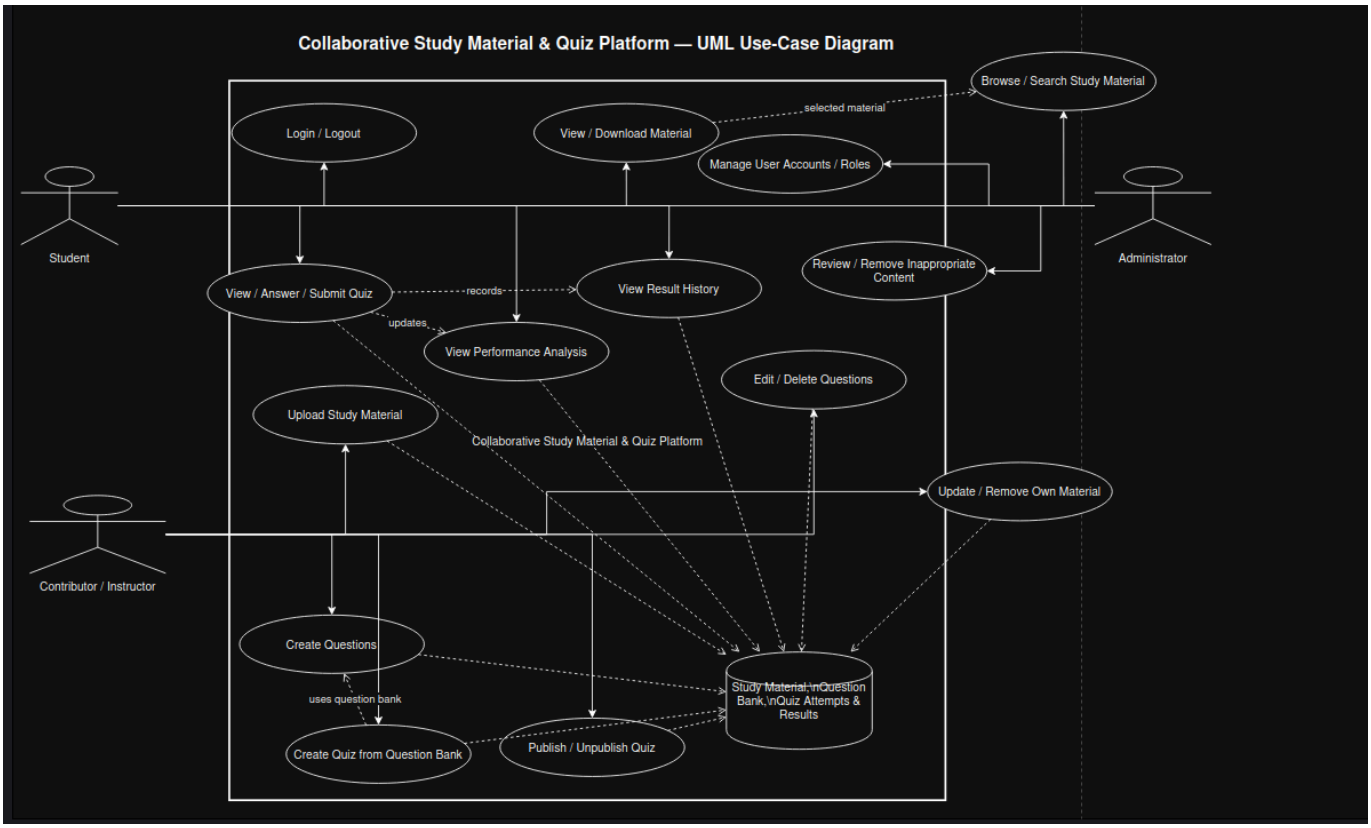
Req ID	Requirement	Priority	Acceptance Criteria / Test Case
CSMQ-SR-001	The system shall require valid authentication before granting access to protected pages or operations.	High	Unauthenticated access is rejected or redirected to login. Test: TC-Sec-01
CSMQ-SR-002	The system shall enforce role-based authorization for Student, Contributor/Instructor and Administrator operations.	High	Unauthorized role attempts are denied. Test: TC-Sec-02
CSMQ-SR-003	The system shall store passwords using a secure one-way password hashing mechanism and shall not store plaintext passwords.	High	Inspection confirms passwords are stored only as hashes. Test: TC-Sec-03
CSMQ-SR-004	The system shall use HTTPS for client-server communication in production.	High	Application traffic is served through HTTPS. Test: TC-Sec-04
CSMQ-SR-005	The system shall validate uploaded files and user-provided inputs before storing or processing them.	High	Invalid file types, oversized files and invalid fields are rejected safely. Test: TC-Sec-05

6. Quality Attributes & Acceptance Tests

The system will be considered acceptable when all high-priority functional requirements pass their tests, no critical security requirement fails, quiz scores and stored results remain consistent, and the stated performance, usability and compatibility targets are met.

- Authentication and role authorization
- Study-material upload, search and access
- Question-bank and quiz creation
- Quiz attempt and automatic evaluation
- Result storage and performance analysis
- Administration and content moderation
- Security, performance and compatibility

7. System Models and Diagrams



8. Requirements Traceability Matrix (RTM)

Req ID	Short Description	Section	Module	Test Case	Status
CSMQ-F-001	The system shall allow a registered user to l...	4.1	Authentication & User Management	TC-Auth-01	N
CSMQ-F-002	The system shall allow an authenticated user ...	4.1	Authentication & User Management	TC-Auth-02	N
CSMQ-F-003	The system shall assign users a role and rest...	4.1	Authentication & User Management	TC-Auth-03	N
CSMQ-F-004	The system shall allow authorized users to up...	4.2	Study Material Management	TC-Mat-01	N
CSMQ-F-005	The system shall allow users to browse and se...	4.2	Study Material Management	TC-Mat-02	N
CSMQ-F-006	The system shall allow authorized contributor...	4.2	Study Material Management	TC-Mat-03	N
CSMQ-F-007	The system shall allow users to view or downl...	4.2	Study Material Management	TC-Mat-04	N

CSMQ-F-008	The system shall allow authorized contributor...	4.3	Question Bank & Quiz Management	TC-QB-01	N
CSMQ-F-009	The system shall allow contributors to edit o...	4.3	Question Bank & Quiz Management	TC-QB-02	N
CSMQ-F-010	The system shall allow contributors to create...	4.3	Question Bank & Quiz Management	TC-Quiz-01	N
CSMQ-F-011	The system shall allow a contributor to publi...	4.3	Question Bank & Quiz Management	TC-Quiz-02	N
CSMQ-F-012	The system shall allow a student to view the ...	4.3	Question Bank & Quiz Management	TC-Quiz-03	N
CSMQ-F-013	The system shall automatically evaluate submi...	4.4	Results & Performance Analysis	TC-Result-01	N
CSMQ-F-014	The system shall store each completed quiz at...	4.4	Results & Performance Analysis	TC-Result-02	N
CSMQ-F-015	The system shall allow students to view their...	4.4	Results & Performance Analysis	TC-Result-03	N
CSMQ-F-016	The system shall provide performance summarie...	4.4	Results & Performance Analysis	TC-Perf-01	N
CSMQ-F-017	The system shall allow an administrator to vi...	4.5	Administration	TC-Admin-01	N
CSMQ-F-018	The system shall allow an administrator to re...	4.5	Administration	TC-Admin-02	N
CSMQ-NF-001	Common operations such as login, material sea...	5	Performance	TC-Perf-01	N
CSMQ-NF-002	The system shall maintain data consistency be...	5	Data Integrity	TC-INT-01	N
CSMQ-NF-003	The interface shall use a consistent layout a...	5	Usability	TC-UX-01	N
CSMQ-NF-004	The application shall be organized into modul...	5	Maintainability	Code Review	N
CSMQ-NF-005	The system shall be accessible	5	Compatibility	TC-Compat-01	N

	through the tw...				
CSMQ-NF-006	The system shall remain available during norm...	5	Reliability	Operations Review	N
CSMQ-SR-001	The system shall require valid authentication...	5.1.2	Security	TC-Sec-01	N
CSMQ-SR-002	The system shall enforce role-based authoriza...	5.1.2	Security	TC-Sec-02	N
CSMQ-SR-003	The system shall store passwords using a secu...	5.1.2	Security	TC-Sec-03	N
CSMQ-SR-004	The system shall use HTTPS for client-server ...	5.1.2	Security	TC-Sec-04	N
CSMQ-SR-005	The system shall validate uploaded files and ...	5.1.2	Security	TC-Sec-05	N

8. Contributions

Neeraj R Rugi - (PES2UG24CS411) Section 1, Section 2

Pradhyumn Anand - (PES2UG24CS358) Section 3, Section 4

Neha Prabhu - (PES2UG24CS312) Section 5, Section 6

Nisarga - (PES2UG24CS325) Section 7, Section 8